

ECOS 3025 25S2

Final review 模型串讲1

Asymmetry information

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# A quick glance of basic definition

- ❖ asymmetric information → 🤔 I know more than you do 信息不对称
- ❖ adverse selection → 🤔 I know my “type”, but you don’t and I may exploit this information to my advantage 逆向选择
- ❖ moral hazard → 🤔 I observe my actions, but you don’t ... and this may be costly for you

1. Lemon market 柠檬市场🍋

2. Signaling in the labor market 信号传递模型

3. principle and agent problem 委托代理问题

# A quick glance



## Adverse selection 逆向选择

- Situation where individuals have hidden characteristics and in which a selection process results in a pool of individuals with undesirable characteristics.
- 逆向选择的**核心机制**是：信息不对称导致市场无法区分高质量与低质量产品，从而让**高质量一方受到不公平的平均定价**。
- Example:
  - people who buy life insurance know more about their own health than does the insurance company → Who is more likely to buy life insurance?
- Adverse selection **reduces size of a market and possibly eliminates it (market failure).**市场失灵
- 信息不对称的结果！！

## Means of equalizing information

- **Screening 筛选**
  - action taken by an uninformed person to determine info possessed by informed people
  - Example: used-car market → potential buyer test drives used cars and take to mechanics for inspection
- **Signaling 信号**
  - action taken by an informed person to send information to a less-informed person
  - Example: product quality → firm distributes a favorable report on its product by an independent testing agency to prove high quality

# Today 's agenda 1: Lemon market 柠檬市场🍋

# Lemons market



信息不对称问题的根源：

- buyer cannot judge a product's quality before purchasing it, while seller knows quality from experiencing the good
- 消费者无法识别市场上商品的质量，因此愿意支付的价格在高和低质量商品的wtp之间

信息不对称问题的结果：

- Owners of low-quality cars are more likely to sell their cars, leading to **adverse selection in used-car markets**. 二手车逆向选择
- 劣币驱逐良币： Low-quality cars (Lemons) may drive high quality ones (Plums) out of the market



# To begin with



## Basic set up of the model:

- used-car market, where two types of cars: high and low quality cars
- many potential buyers for used cars, all are willing to pay:

\$1,000 for a lemon

\$2,000 for a good used car

- Sellers: 1,000 lemons and 1,000 good used cars
- reservation price of owners :

\$750 for lemons

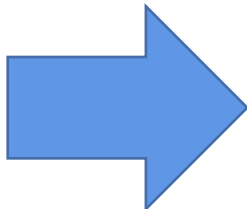
\$1,250 or \$1,750 for good cars

| Type of Car | Probability | Buyers' willingness to pay | Sellers' reservation price |
|-------------|-------------|----------------------------|----------------------------|
| Lemons      | 0.5         | \$1000                     | \$750                      |
| Plums       | 0.5         | \$2000                     | \$1250 or \$1750           |

# Case 1 : what if perfect information?

Case 1 完美信息: (both buyers and sellers can identify quality)

| Type of Car | Probability | Buyers' willingness to pay | Sellers' reservation price |
|-------------|-------------|----------------------------|----------------------------|
| Lemons      | 0.5         | \$1000                     | \$750                      |
| Plums       | 0.5         | \$2000                     | \$1250 or \$1750           |



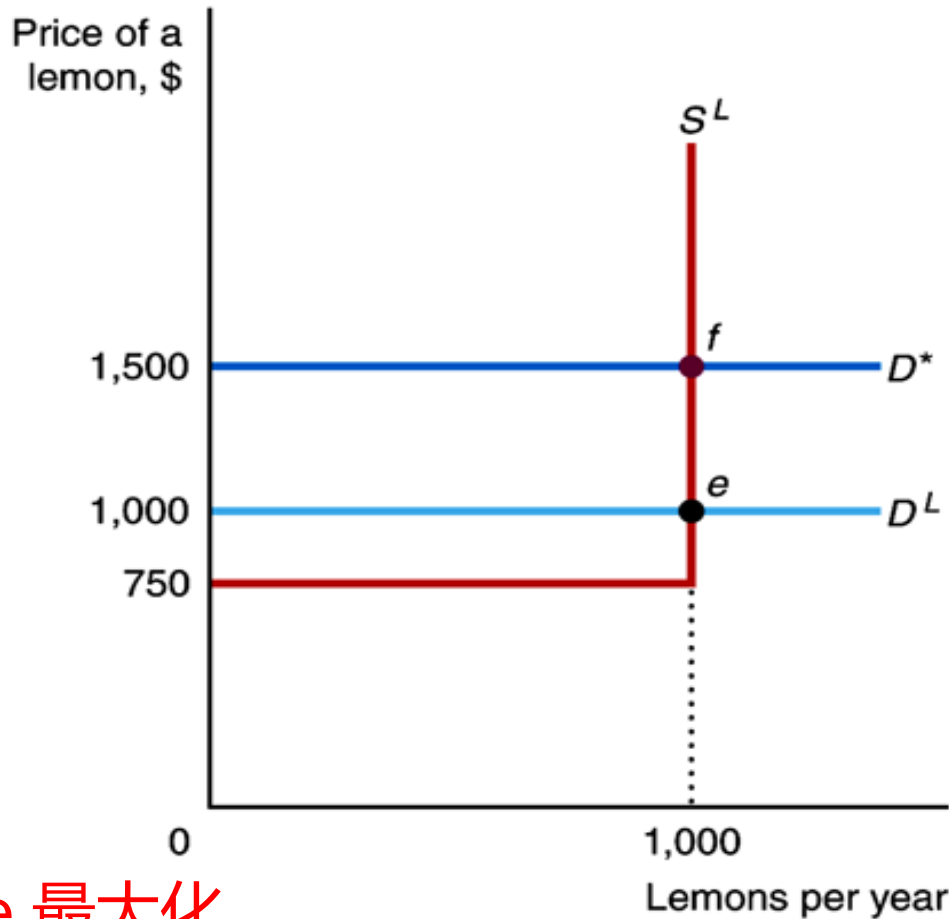
Separated market : Supply =demand !

- Plums sell for 2000
- Lemons sell for 1000
- Market is **efficient**: the cars go to people who value them the most
- Points ‘e’ and ‘E’ on the graph

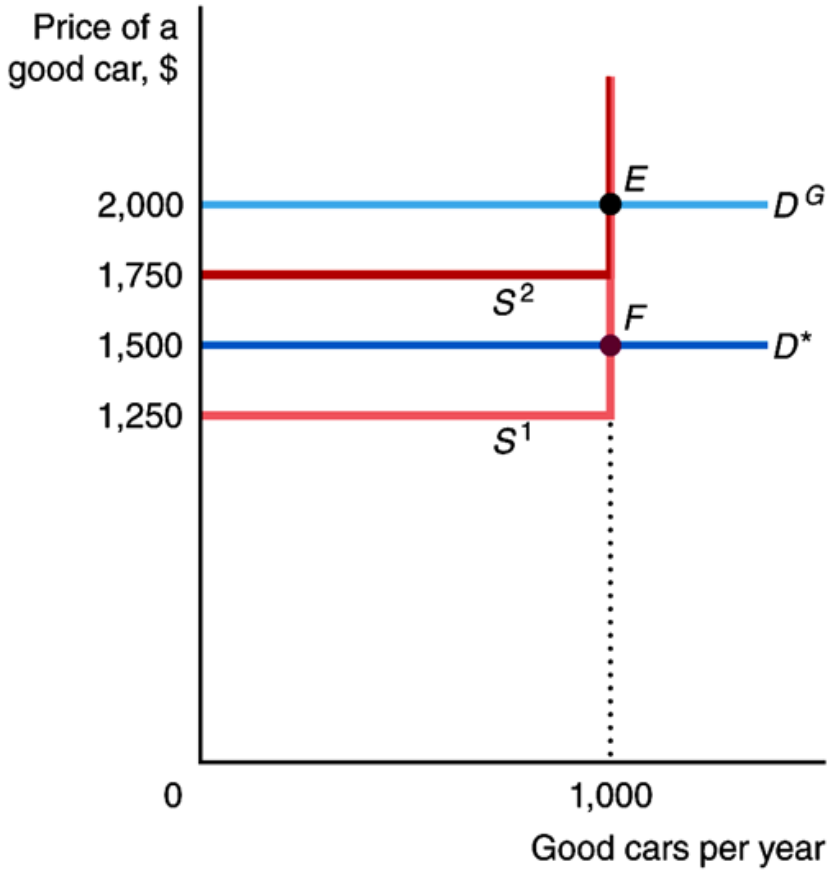
⚠️Efficiency: 关注quantity sold的量, 而不是who gain more from the trade ---social welfare 最大化

只要social benefit >social cost ,就应该生产&consume!

(a) Market for Lemons



(b) Market for Good Cars



# Case 2 : what if imperfect information?

Case 2 不完美信息但对称! : Information is *imperfect* but *symmetric*

- ! both buyers and sellers cannot identify quality!!

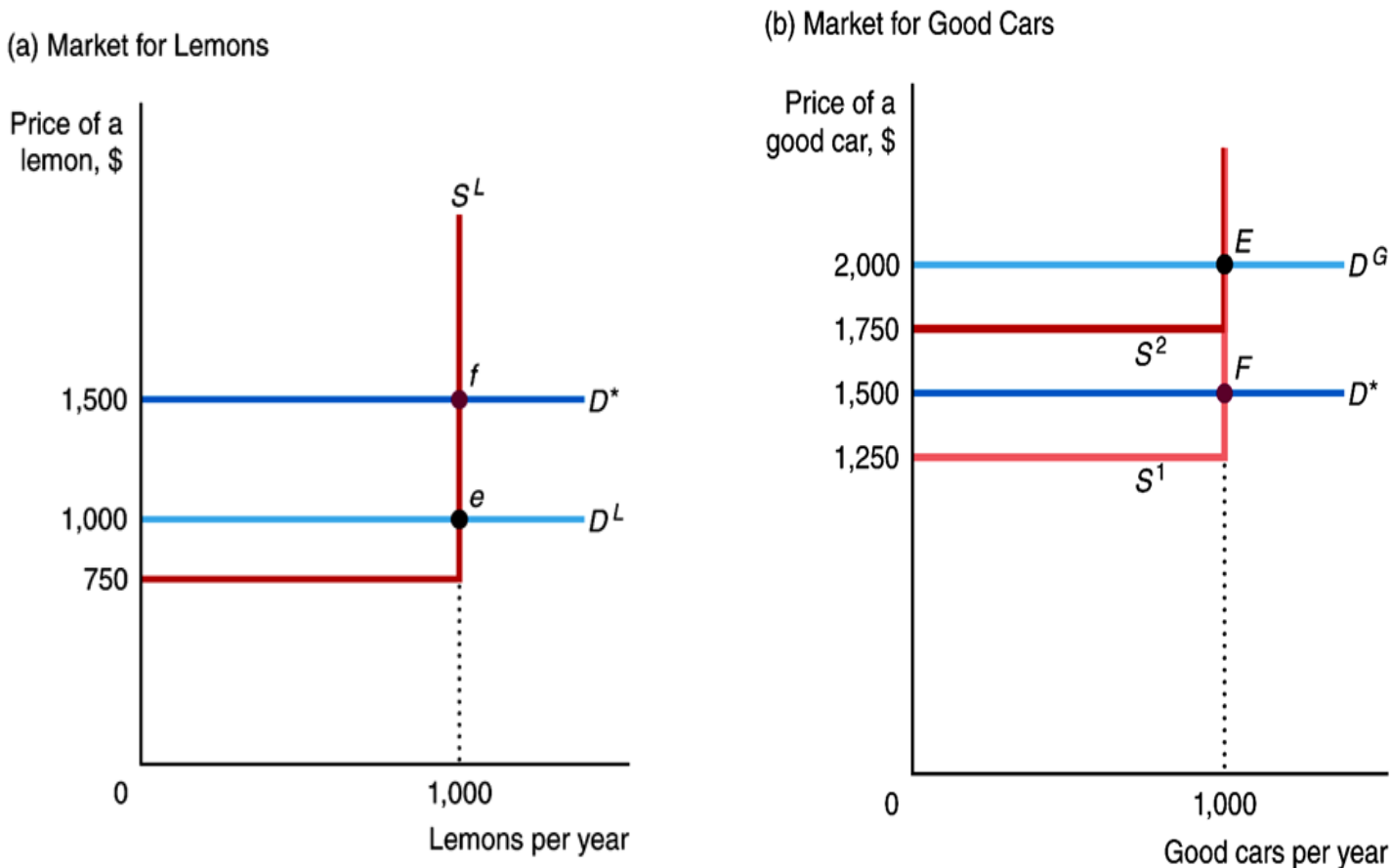
**Expected WTP: (Risk-neutral)** buyers are willing to pay based on expected value:  $(0.5 \times 1000) + (0.5 \times 2000) = 1500$

**Expected reservation price:** Sellers (with  $v = 1250$  or  $1750$ ) will take  $(0.5 \times 750) + (0.5 \times v) = (375) + (0.5 \times v) < 1500$

All cars sell for 1500

One pooled market

Market is **efficient!**



| Type of Car | Probabilit<br>y | Buyers'<br>willingness to<br>pay | Sellers'<br>reservation price |
|-------------|-----------------|----------------------------------|-------------------------------|
| Lemons      | 0.5             | \$1000                           | \$750                         |
| Plums       | 0.5             | \$2000                           | \$1250 or \$1750              |

# Case 3 : what if asymmetric information?



## Case 3.1 信息不对称! : Information is *asymmetric*

- ! sellers know the type but buyers don't! !
- Assume :  $v=1250$  seller's reservation price ,如果plum 没有那么 “贵”
- ❖ Buyers willing to pay 1500 (since they don't know type), expected WTP
- ❖ Sellers of both types of cars are willing to accept 1500 (plum 也愿意接受, 虽然可能心理 “不平衡” 😊)
- ❖ All cars sold for 1500, Market is efficient
- ❖ ! (asymmetric information hasn't caused inefficiency) 没有导致social welfare的减少!

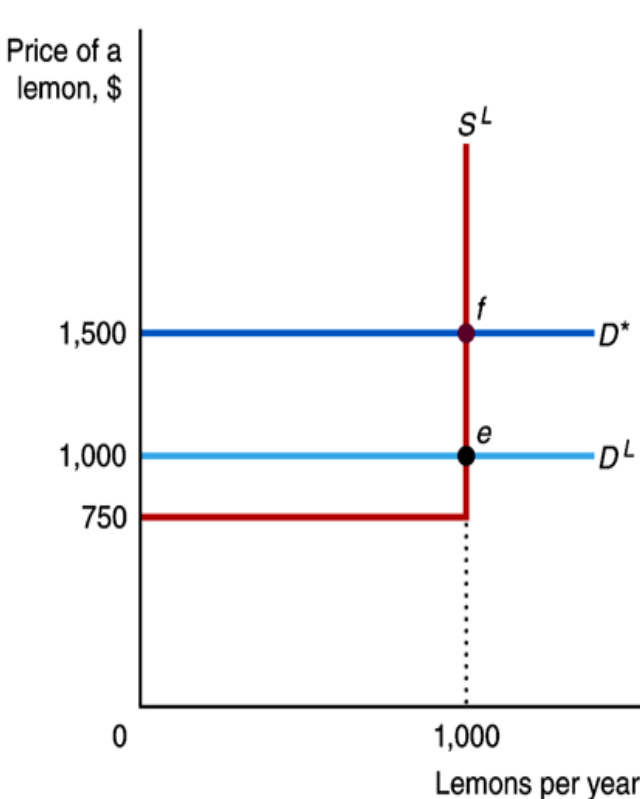
只是更多的surplus 流向了lemon sellers: Sellers of Lemons **benefit** while sellers of Plums **suffer**

Points ‘f’ and ‘F’ on the graph

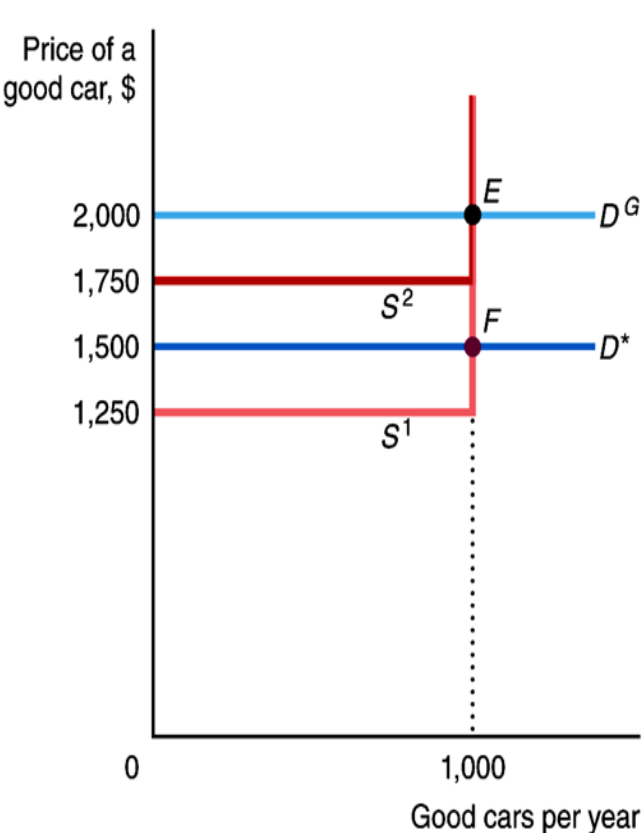
Buyers would be lucky to get a good bargain (plum) or unlucky to end up with a lemon

| Type of Car | Probability | Buyers' willingness to pay | Sellers' reservation price |
|-------------|-------------|----------------------------|----------------------------|
| Lemons      | 0.5         | \$1000                     | \$750                      |
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(a) Market for Lemons



(b) Market for Good Cars



💡 Efficiency: 关注quantity sold的量, 而不是who gain more from the trade ---social welfare 最大化

# Case 3 : what if asymmetric information?

## Case 3.2 信息不对称! : Information is *asymmetric*

- ! sellers know the type but buyers don't! !
- Assume :  $v=1750$  seller's reservation price ,如果plum 非常 “贵”
  - ❖ Buyers willing to pay 1500 (since they don't know type), expected WTP
  - ❖ Sellers of plums not willing to sell for 1500
  - ❖ Buyers know that only cars they can buy at  $p < \$1,750$  are lemons, **WTP =1000**
  - ❖ ! (asymmetric information caused inefficiency) **Only Lemons will be sold!**

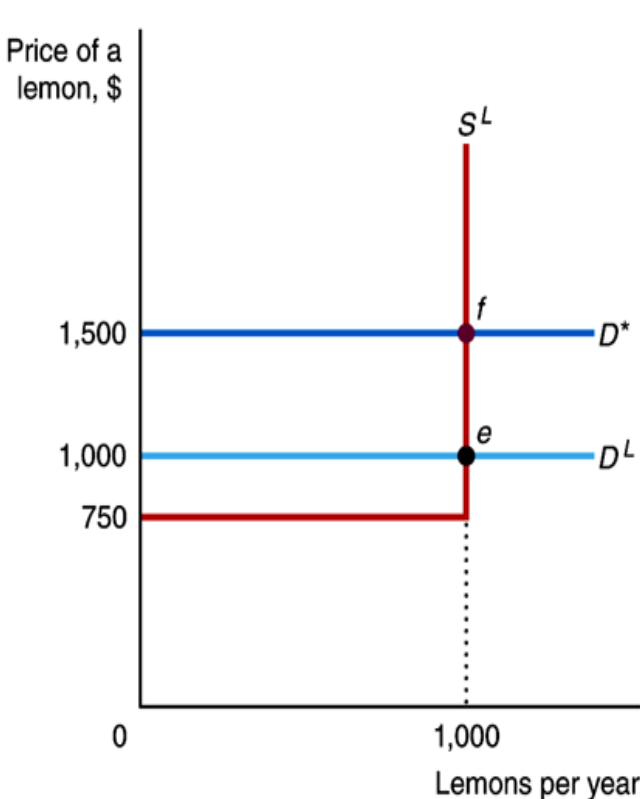
劣币驱逐良币: **Lemons drive good cars out of the market - The ‘Lemons problem’**

Point ‘e’ on the graph (only one point)

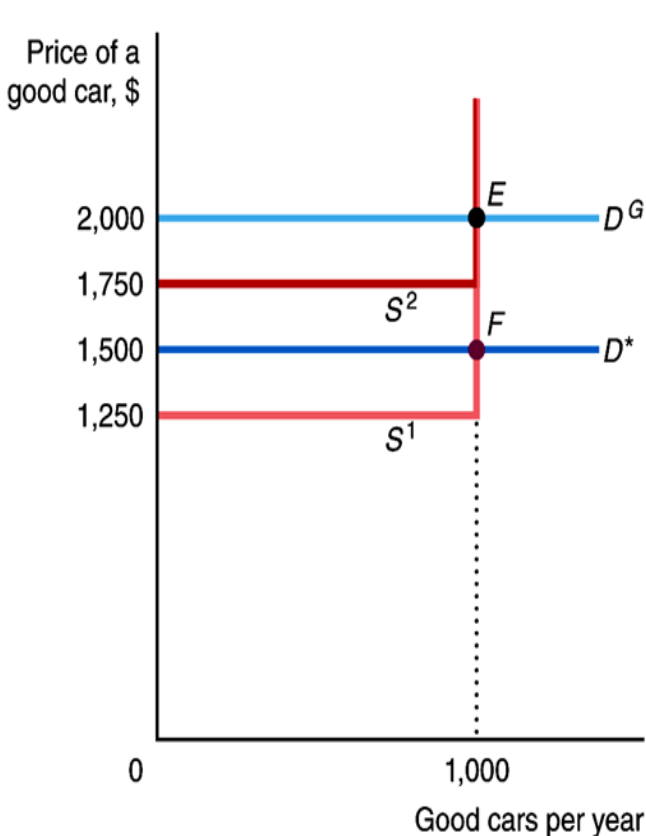
- ❖ Market **is inefficient**: Buyers willing to pay 2000 for plum and sellers accept 1750, but the trade does **not** happen. 社会总福利减少! !

| Type of Car | Probability | Buyers' willingness to pay | Sellers' reservation price |
|-------------|-------------|----------------------------|----------------------------|
| Lemons      | 0.5         | \$1000                     | \$750                      |
| Plums       | 0.5         | \$2000                     | <b>\$1750</b>              |

(a) Market for Lemons



(b) Market for Good Cars



# Lemon case with continuous quality

The quality of a car is described by its “type”  $t$ .  $t$  is *uniformly* (evenly) distributed between 0 and 2000.

$$t \sim U[0, 2000]$$

payoffs:

$$\begin{aligned} \text{(buyer)} \quad U^B(t, p) &= \begin{cases} bt - p & \text{if there is a sale} \\ 0 & \text{otherwise} \end{cases} \\ \text{(seller)} \quad U^S(t, p) &= \begin{cases} p - t & \text{if there is a sale} \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

where  $b > 1$

The seller knows  $t$ , but the buyer does not. Will trade occur?

- ❖ For seller to sell  $U^S(t, p) = p - t \geq 0$
  - ❖ ---卖价一定不小于quality! 而且buyer知道!!
  - ❖ 质量uniformly均匀分布, reservation price也均匀分布, 成本→价格原则, 生产上的愿意卖的最低价格也会均匀分布!
- Therefore, the buyer knows that
- $t \leq p$
  - conditional on sale, average quality will be  $p/2$
- ❖ For buyer to buy:  $U^B(t, p) = bp/2 - p \geq 0 \Rightarrow b \geq 2$
  - ❖ the buyer **must value the car much more than the seller!!**

adverse selection leads to less trade

## **agenda 2:** Signaling in the labor market 信号传递模型

# Asymmetric information: Hiring and Cheap talk



Asymmetric information creates problems in labor markets.

Worker *signaling* and firm *screening* may reduce problems.

people can use **cheap talk** to distinguish themselves or their attributes at low cost.

Cheap talk: **unsubstantiated claims or statements** “无证据随便说”

Case study: Potential employee knows his/her ability (**H or L**), which is **unobservable to the firm**.

Two kinds of jobs: firm prefers **H workers in hard job, L in easy job**.

**Cheap talk works: employee can claim H or L, and is believed by employers.**

- Case 1 Cheap talk may reveal employees ability correctly.
- Case 2: Cheap talk may not work: worker claims H regardless of true type

| Payoff matrix    |   | Firm assigns worker |          |
|------------------|---|---------------------|----------|
|                  |   | Hard job            | Easy job |
| Workers' ability | H | 3, 2                | 1, 1     |
|                  | L | 1, 1                | 2, 4     |

| Payoff matrix    |   | Firm assigns worker |          |
|------------------|---|---------------------|----------|
|                  |   | Hard job            | Easy job |
| Workers' ability | H | 3, 2                | 1, 1     |
|                  | L | 3, 1                | 2, 4     |

- Firm assigns job on the basis of expected value: only easy job assigned  
Expectedly, hard job < easy job
- Inefficient if worker is H type: **做hard job可以给社会产生更多价值**  
When cheap talk is not credible, an individual may have to send an expensive signal (through education for instance).

# Asymmetric information: signaling



## Case study:

Two types of workers: H (high ability) and L (low ability).

$\theta$  is the proportion of H individuals.

Value of H worker is  $W_H$  and value of L worker is  $W_L$ .

Only H workers can get **education** at the cost of  $C$ , which is **too costly for L workers**.

Case 1 without signaling: not possible to distinguish between types,

- **Pooling equilibrium:** everyone is paid the average wage at  $WP = \theta W_H + (1 - \theta) W_L$ ---- expectation

Case 2 with education as signal:

- **Separating equilibrium:** H go to university and get a degree.

H get paid  $W_H$ , L get paid  $W_L$

This outcome prevails if  $W_H - C > W_L$ : **low skill 读书的代价太大, 不可能选择读书, 因此如果high选择读书, 则可separate!**

- **Pooling equilibrium:** H does not get a degree if  $W_H - WP < C$ . 如果high 选择不读书, 则就算市场上education是筛选的有效信号, 也无法筛选

# Asymmetric information: signaling



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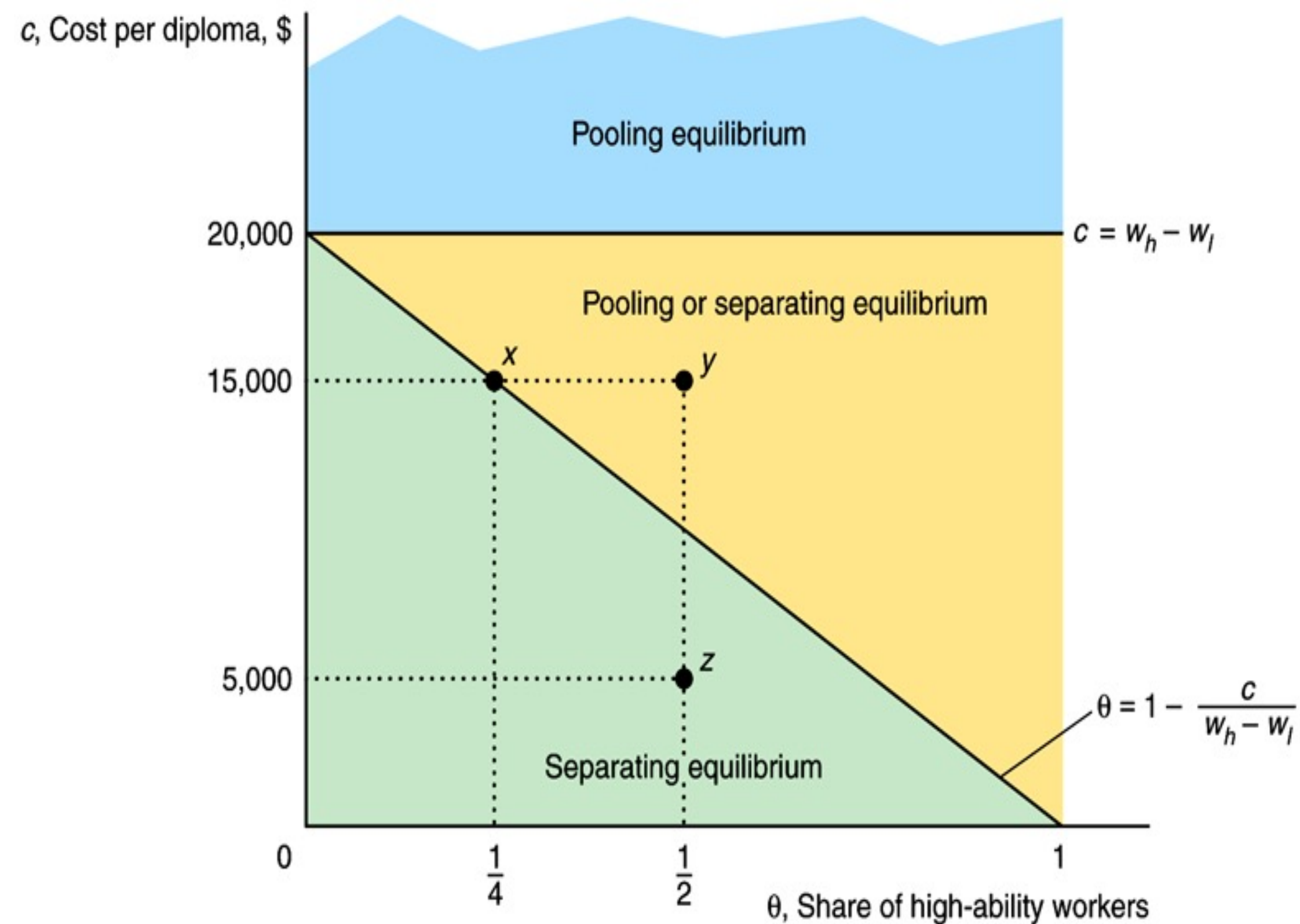
❖ **就算读书成本很低，  $W_H - W_P > C$  , H may not get a degree if  $W_H - C < E(W) = W_P = \theta W_H + (1 - \theta) W_L$**

-----high skill 占比很大， expected wage很高，与之相比读书成本太高

----- less likely that they will wish to signal. 依旧可能是pool

# Asymmetric information: signaling

$$W_H = \$40,000 \text{ and } W_L = \$20,000$$



❖ H does not get a degree if  $W_H - W_P < C$ . 如果high 选择不读书，则就算市场上 education 是筛选的有效信号，也无法筛选

❖ 就算读书成本很低，  $W_H - W_P > C$  , H may not get a degree if

$$W_H - C < E(W) = W_P = \theta W_H + (1 - \theta) W_L$$

-----high skill 占比很大， expected wage 很高， 与之相比读书成本太高

-----less likely that they will wish to signal. 依旧可能是pool

$$W_H - W_L = 20,000 \quad (1)$$

$$W_H - c = 40,000 - c \quad (2)$$

$$E(w) = \theta W_H + (1 - \theta) W_L \quad (3)$$

$$(2)=(3): \quad \theta = 1 - \frac{c}{W_H - W_L}$$

# Asymmetric information: signaling

Signal 本身也是一种资源的浪费: no increase in productivity

- only a signal and does not effect ability
- Efficiency could be increased by banning signaling.
- Signaling makes H workers earning more than L workers. It is possible for everyone to be worse off in a separating equilibrium. This is because of the **inefficiencies arising from the need for signaling due to the asymmetric information about underlying abilities.**

因signal 而额外浪费的资源会通过成本-价格传递链条收到影响, 每个role 都 worse off

因此, 从**social welfare 角度: pooling is more welfare efficient**

## agenda 3: principle and agent problem 委托代理问题

❖ moral hazard

🤔 I observe my actions, but you don't and this may be costly for you

# principle and agent problem

## Case study:

A firm hires a manager who can **exert “high” or “low” effort** ( $e=1$ ,  $e=0$ )

Firm cannot monitor manager effort.

Firm revenue depends on **manager effort and on the state of the economy.**

**Cost ( $e=0$ )=0  $\Rightarrow$  \$20K (boom) and \$10K (bust)**

**Cost ( $e=1$ )=\$10K  $\Rightarrow$  \$40K (boom) and \$20K (bust)**

**the probability of a boom or bust =  $\frac{1}{2}$ .**

## ❖ **Manager’s (Agent’s) Problem:**

Choose an effort level ( $e=0,1$ ) in order to **maximize her payment, net of the cost of effort.**

## ❖ **Owner’s (Principal’s) Problem:**

Choose the wage to pay the manager in order to **maximize expected profit**

# principle and agent problem



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**the probability of a boom or bust =  $\frac{1}{2}$ .**

## Contract Options:

❖ **Owner pays manager a fixed wage of \$5K.** 固定工资

❖ **Owner offers manager following performance contract:** 工资与绩效相关，但固定，不**share** 利润

If Rev=\$10K or \$20K  $w=0$

If Rev=\$40K  $w=\$24K$

❖ **Owner offers manager the following revenue sharing contract:** 工资**share** 利润

If Rev>\$18K  $w=\text{Rev}-\$18K$

If Rev≤\$18K  $w=0$ .

# 总结：moral hazard 和adverse selection 时间顺序转化问题



- Adverse selection: 购买前, 担心质量, 优质商家为了吸引消费者高价购买: warranties and guarantees
- Moral hazard: 购买后, “摆烂”使用; warranties and guarantees can introduce moral hazard issues because now the buyers no longer bear the risk and so may use products harder, more often or inappropriately to the intended use.
- **双重问题的解决方案: 提高保修的“兑付”门槛: not valid except under very clearly specified conditions.**

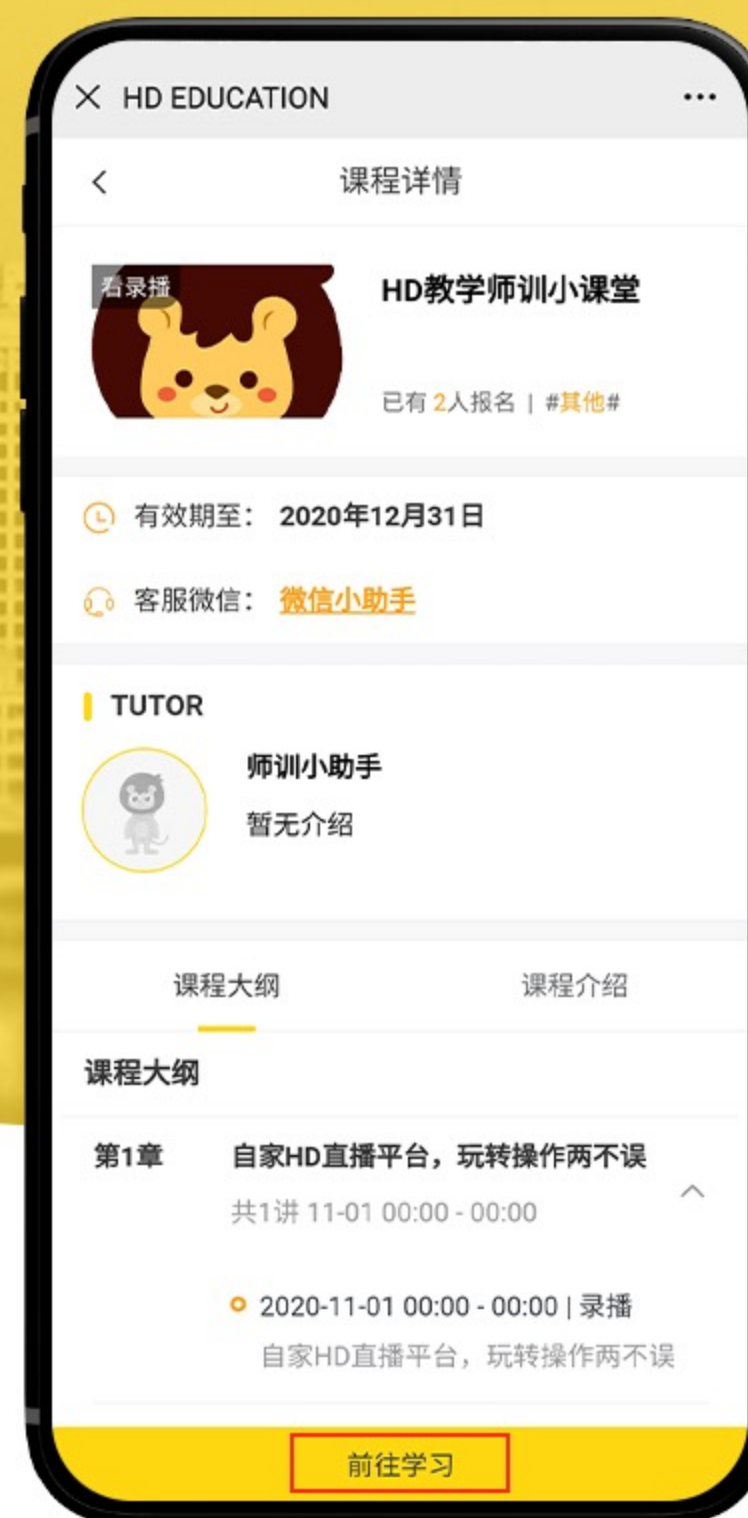
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